

CUSTOMER CHURN ANALYSIS

Analyst Assignment Brief — Questions & Dashboard Specifications

You have been engaged to investigate why our customers are discontinuing their relationship with the company. The dataset provided covers customer demographics, subscribed services, billing details, and churn status. Your deliverables are structured below across three sections: **Excel Analysis**, **SQL Analysis**, and the **Executive Dashboard Specification**.

SECTION 1 | Excel Analysis Questions

- Q 1** Using a Pivot Table, break down the churn rate (%) by Contract Type (Month-to-month, One Year, Two Year). Which contract segment drives the highest volume of churned customers?
- Q 2** Create a tenure-band column (e.g., 0–12 months, 13–24 months, 25–48 months, 49+ months) using IF/IFS formulas. Then, using a Pivot Table, calculate the average MonthlyCharges and average TotalCharges for churned vs. retained customers within each tenure band.
- Q 3** Using COUNTIFS and AVERAGEIFS, compute the churn rate for customers who have NO OnlineSecurity AND NO TechSupport simultaneously. Compare this rate against customers who have BOTH services active.
- Q 4** Build a matrix Pivot Table with InternetService type (DSL, Fiber Optic, No) on rows and PaymentMethod on columns. Show the count of churned customers in each cell. Apply Conditional Formatting (heat map) to highlight the highest-risk combinations.
- Q 5** Using the CORREL function, measure the correlation between MonthlyCharges and Churn (encode Churn as 1/0). Additionally, create a scatter plot of MonthlyCharges vs. tenure colour-coded by Churn status to visually identify the high-risk billing zone.

SECTION 2 | SQL Analysis Questions

- Q 1** Write a SQL query using window functions (RANK or DENSE_RANK) to identify the top 3 PaymentMethod and Contract Type combinations ranked by churn rate (%). Include the total customer count and churned customer count for each combination.
- Q 2** Using a CTE (Common Table Expression), segment customers into three MonthlyCharges tiers (Low: <\$40, Mid: \$40–\$70, High: >\$70). For each tier, calculate the churn rate, average tenure, and the percentage contribution of each tier to total churn.

Q 3 Write a query to calculate the month-over-month churn cohort: group customers by their tenure bucket (every 6 months) and return the churn rate, the most common InternetService type, and the most common Contract type for each cohort using GROUP BY and subqueries or CTEs.

Q 4 Identify customers who have BOTH StreamingTV = 'Yes' AND StreamingMovies = 'Yes' but NO OnlineSecurity and NO TechSupport. Among this segment, calculate the churn rate and average MonthlyCharges, and compare against all other customers using a single SQL query with CASE WHEN grouping.

Q 5 Using a self-join or subquery, flag each customer as 'High Risk' if their MonthlyCharges exceed the average MonthlyCharges of churned customers in their respective InternetService category. Return the count of High Risk customers by SeniorCitizen status (0/1) and their overall churn rate within that risk flag.

SECTION 3 | KPI Cards for the Executive Dashboard

Overall Churn Rate (%) Primary headline metric — % of total customers who churned	Total Customers Lost Absolute count of churned customers in the period	Avg. Tenure of Churned Customers In months — reveals how early in the lifecycle we lose customers
Avg. Monthly Charges (Churned vs. Retained) Side-by-side comparison to signal price sensitivity	Revenue at Risk (\$) Monthly revenue lost = churned customers x their avg. MonthlyCharges	Churn Rate — Month-to- Month Contracts Spot the highest-risk contract segment instantly

SECTION 4 | Dashboard Charts & Visualisations

#	Chart Title	Chart Type	Business Purpose
0 1	Churn Rate by Contract Type	<i>Clustered Bar Chart</i>	Compare churn % across Month-to-Month, One Year, and Two Year contracts — the single biggest lever.
0 2	Monthly Charges Distribution: Churned vs. Retained	<i>Box Plot / Violin Chart</i>	Expose the price band where churn risk spikes. Reveals whether high-bill customers leave more.
0 3	Churn Rate by Tenure Band	<i>Line / Area Chart</i>	Track how churn probability evolves over the customer lifecycle. Identify the critical early-exit window.
0 4	Service Adoption vs. Churn Heatmap	<i>Heatmap Matrix</i>	Rows = services (OnlineSecurity, TechSupport, StreamingTV...), Columns = Churn/Retained. Shows which service gaps drive exits.
0 5	Churn by Internet Service & Payment Method	<i>Stacked Bar / Grouped Bar</i>	Pinpoint the riskiest product-payment combinations for targeted retention campaigns.

06	Senior Citizen Churn Breakdown	<i>Donut / Pie Chart</i>	Proportion of churned customers who are senior citizens — flags a demographic needing tailored outreach.
07	Top Churn Risk Segments	<i>Treemap</i>	Size = volume of churned customers; segments defined by Contract × InternetService × Tenure Band.